P. Swathika

8870700760 | swathika1911@gmail.com | LinkedIn | Github | Leetcode

OBJECTIVE

Results-driven 3rd-year B.Tech student specializing in Artificial Intelligence and Data Science with hands-on internship experience in AI, Machine Learning, and full-stack web development. Demonstrated research excellence with 85%-90% model accuracy across multiple published projects. Skilled in Python, ML frameworks, and image processing, with a passion for developing scalable, intelligent systems

EDUCATION

- **Bachelor of Technology** – Artificial Intelligence & Data Science | 2023 - Present
  SIMATS Engineering, Chennai, Tamil Nadu
  CGPA: 8.9 / 10.0
- Higher Secondary Certificate (HSC) | 2023 | Percentage : 86%
  Laurel Higher Secondary School, Adirampattinam.

SKILLS

- **Programming Languages:** Python, Java, C, R
- **Web Technologies:** HTML5, CSS3, JavaScript
- **Libraries & Frameworks**: NumPy, Pandas, Scikit-learn, OpenCV, TensorFlow
- **Tools & Platforms:** GitHub, Visual Studio Code
- **Core Concepts:** Data Structures & Algorithms, Machine Learning, DBMS, Operating Systems

INTERNSHIP EXPERIENCE

Artificial Intelligence Intern | L&T Shipbuilding, Chennai | 2025

- Engineered an AI-powered digital library system for SCM, automating document classification & retrieval workflows.
- Reduced document search time by ~40% using intelligent indexing and NLP-based retrieval techniques.

Machine Learning Intern | Gateway Software Solutions, Coimbatore | 2025

- Built & deployed an ML-based document management system to streamline storage, indexing & retrieval processes
- Improved query accuracy and operational efficiency using AI-driven search and indexing techniques.

PROJECTS

SharpX – Shape Detection App | Python, OpenCV, Image Processing

- Developed a real-time geometric shape recognition system using contour and edge detection techniques in OpenCV.
- Optimized the image processing pipeline to improve detection speed and accuracy across varied inputs.

Notice Board Portal | Python

- Built a web-based digital notice board with a secure admin interface for real-time content creation and updates.
- Replaced manual communication with a centralized system, improving information accessibility and efficiency.

Jeejee Garlands – E-Commerce Frontend | HTML5, CSS3, JavaScript

- Developed a responsive, mobile-first frontend for a traditional garland and saree service platform.
- Designed structured product and service pages to enhance user navigation and overall user experience

RESEARCH

Smart Farming Optimization using ANN & RNN | Tech Star Summit 2024 – SIMATS Engineering

Designed and trained ANN & RNN models on agricultural datasets, achieving ~91.75% accuracy with RNN outperforming ANN. Applied T-test validation and proposed an AI-based framework for precision agriculture and resource optimization.

Crop Mapping using KNN & CNN Algorithms | Tech Star Summit 2024 – SIMATS Engineering

Developed a crop classification pipeline using KNN and CNN, achieving ~94.65% accuracy with KNN on diverse datasets. Conducted comparative ML vs deep learning analysis to evaluate efficiency and improve agricultural insights.

Hematopoietic Allogeneic Blood Transfusion – ML Analysis | Tech Star Summit 2025 – SIMATS Engineering

Performed comparative analysis using Decision Tree, SVM, and KNN to predict blood donor compatibility. Achieved highest accuracy with Decision Tree and optimized SVM for large datasets to reduce transfusion risks.

CERTIFICATIONS

- **NPTEL** - Internet of Things (IoT)
- **Artificial Intelligence Fundamentals** - HP LIFE
- **Python Programming Certification** - GUVI
- **Oracle Database SQL Certification** - Oracle Academy

KEY ACHIEVEMENTS

- Achieved 94.65% (KNN) and 91.75% (RNN) model accuracy in research projects
- Presented 3 research papers at Tech Star Summit
- Completed 2 AI/ML internships delivering real-world solutions